

A Novel Explainable Machine Learning Framework for Credit Risk Prediction with Loan Eligibility and EMI Recommendation

Abstract

The increasing demand for digital banking and online lending has created a need for accurate, reliable, and transparent credit risk assessment systems. Traditional loan approval methods mainly depend on manual verification and fixed credit rules, which are often time-consuming and may lead to incorrect lending decisions. These methods may approve risky borrowers or reject eligible customers because they cannot effectively analyze complex financial information. To overcome these limitations, this research proposes a novel explainable Machine Learning framework for predicting customer loan default risk. The proposed system preprocesses loan and customer financial data, performs feature engineering, and trains different Machine Learning models such as Random Forest, XGBoost, and CatBoost to identify the best-performing model. Explainable Artificial Intelligence techniques, namely SHAP and LIME, are integrated to provide clear explanations for each prediction by identifying the factors influencing loan decisions. Based on the predicted risk level, the system recommends loan eligibility, a suitable loan amount, an affordable EMI, and generates a customer risk score to support better lending decisions. A user-friendly web application is developed to provide real-time loan prediction and decision support. The proposed framework aims to improve prediction accuracy, increase transparency, reduce loan defaults, and assist financial institutions in making faster and more reliable credit decisions.

Keywords: Credit Risk Prediction, Machine Learning, Explainable Artificial Intelligence, XGBoost, CatBoost, SHAP, LIME, Loan Eligibility, EMI Recommendation, Risk Score.

Introduction

Credit risk prediction is an important process in the banking and financial sector because it helps financial institutions determine whether a customer is capable of repaying a loan. Accurate credit risk assessment reduces financial losses, improves the quality of lending decisions, and supports effective credit management. However, traditional loan approval methods mainly rely on manual verification, predefined credit rules, and basic financial indicators. These methods require more

processing time and often fail to capture the complex relationships between customer financial behaviour, repayment history, and creditworthiness. As the number of loan applications continues to increase, there is a growing need for intelligent systems that can automate the credit evaluation process while providing accurate and transparent decisions.

Machine Learning has become an effective solution for predicting loan default risk because it can learn hidden patterns from large financial datasets and produce accurate predictions. Several Machine Learning algorithms, including Random Forest, XGBoost, CatBoost, Logistic Regression, and Decision Trees, have been successfully applied in credit risk assessment. Although these models achieve good prediction performance, many existing studies mainly focus on prediction accuracy and provide limited explanation for their decisions. This lack of transparency makes it difficult for banks to justify loan approval or rejection decisions and reduces user confidence in automated systems. In addition, most existing approaches only classify customers as safe or risky without recommending suitable loan amounts or affordable EMI plans based on customer financial conditions.

To address these limitations, this research proposes The proposed framework combines advanced Machine Learning models with Explainable Artificial Intelligence techniques to improve both prediction accuracy and decision transparency. The system preprocesses customer financial data, performs feature engineering, compares multiple Machine Learning models, and selects the best-performing model using evaluation metrics such as accuracy, precision, recall, and F1-score. Hyperparameter tuning and data balancing techniques are applied to further improve model performance. SHAP and LIME are integrated to explain each prediction by identifying the most influential features affecting the loan decision.

Based on the predicted loan default risk, the proposed system recommends loan eligibility, a suitable loan amount, an affordable EMI, and generates a customer risk score according to the applicant's financial profile and repayment capacity. The complete framework is implemented as a real-time web application that provides instant loan risk prediction and decision support for both customers and financial institutions. By combining accurate prediction, explainable decision-making, and personalized loan recommendations, the proposed system aims to reduce loan defaults, improve lending efficiency, and support fair, reliable, and transparent credit risk assessment.